UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 7A - Work Measurement: Time Studies I

EIN 3241: Human Factors and Ergonomics
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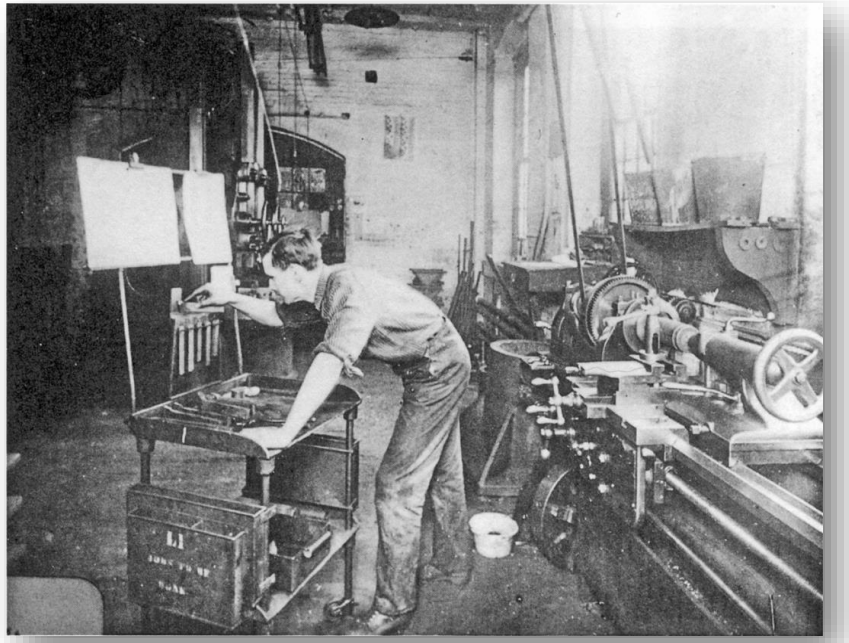
Admin Issues

- Project:
  - Progress Report due TONIGHT!
- Quiz:
  - Wednesday, April 1
  - Lectures 5F, 5G, 6A, and 6B
- Lab 6: Friday, April 3

History of time studies

- Taylor timed how long it took workers to do certain jobs and tried to improve the jobs
- Frank and Lilian Gilbreth later expanded on time studies with time and motion studies



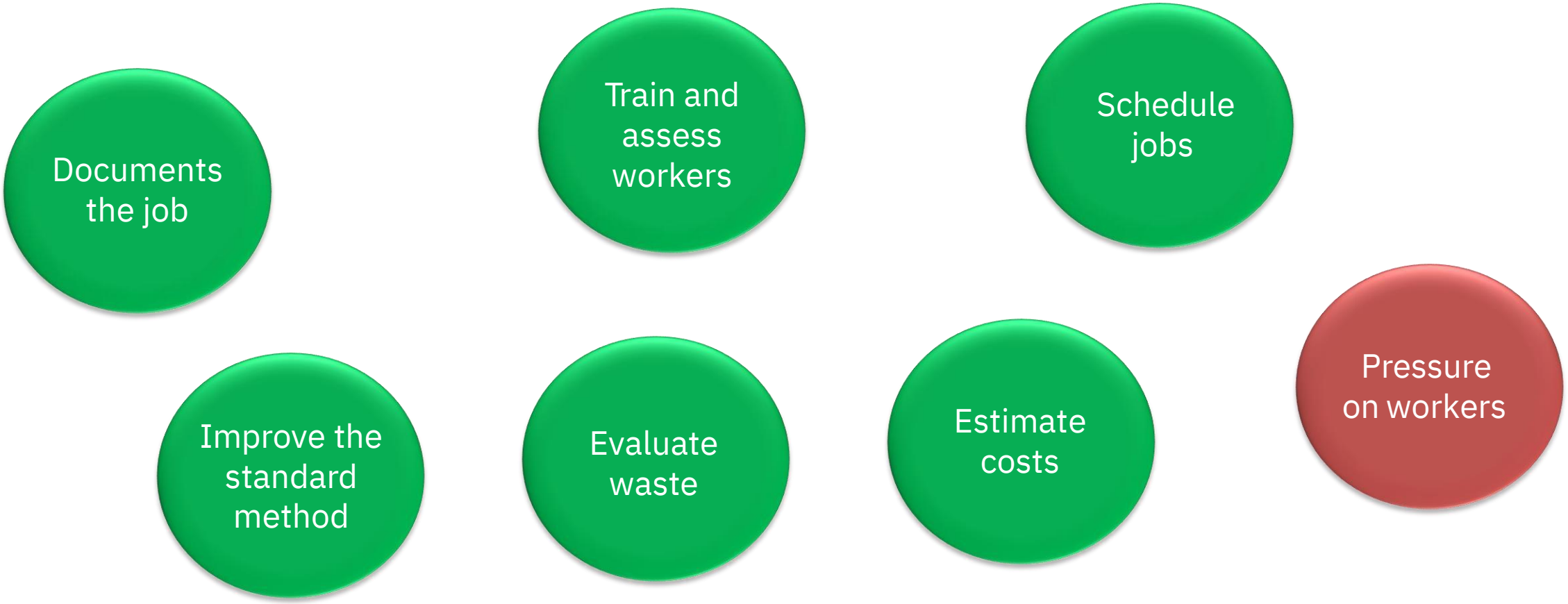
Time study

- Approach to calculating the time to complete a job that relies on breaking a job down into work elements and timing how long each one takes
- Standard time: time required for a qualified worker to perform a job using a standard method under normal conditions
  - Qualified worker: fully trained for the job
- Standard method: the method on which the standard time is based – ideally, the "best" method.
  - Methods are described as a sequence of elements

Can help with calculating standard time

Introduction

- Standard time is calculated after the standard method has been identified



Time study requirements

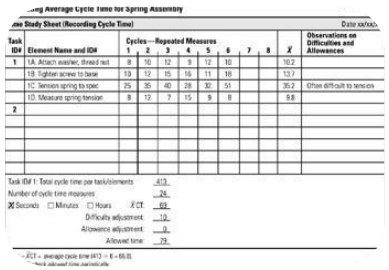
- What types of jobs are best suited for a time study?
  - Job involves manual labor
  - Job involves repetitive cycles
  - Focus on one person

- Operator should be thoroughly familiar with the job (qualified worker)
- Standard method must be used
- Operator and supervisor must know that the job is being studied
  - Need to obtain approval from both
  - Analyst should not be biased and should refrain from any operator criticism

Time study equipment

Historically

Stopwatch



Time study form

Clipboard

Nowadays

Time study software

Video camera

Check out the different apps available (e.g., WorkStudy+)

Time study procedure

1. Identify breakpoints for your elements
2. Determine number of cycles needed
3. Record the element times
4. Address difficulties in the data and clean up data
5. Calculate the elemental times
6. Performance rate the operator
7. Assign appropriate allowances
8. Calculate the standard time

Identify breakpoints for your elements

- Make sure your work elements are not less than
  - 0.04 minutes (~3 seconds)
- Elements might be regular or irregular
  - **Regular element**: happens every cycle
  - **Irregular element**: does not happen every cycle
- To make elements easier to isolate, identify both sight and sound break points.

Note that an abnormal element is not part of the standard method

- Operator deviated from standard method
- Something goes wrong
- Operator had to stop for some reason
- Interference from other workers

Example 1

- Develop the standard method and include breakpoints for each one

Element	Start point	End point
1. Put the batter on the machine and spread	Grasp the butter	Release the wooden compass
2. Wait for crepe to cook	Release the wooden compass	Grasp knife
3. Put ingredients	Grasp knife	Release knife
4. Flip crepe and serve	Release knife	Crepe on plate

Time study procedure

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Number of cycles

- Cannot just rely on one recording of the operation
- How many samples do you need?
  - Use an approximate guide
  - Use statistical methods
- Approximate guide: estimate the number of cycles needed based on the cycle time of the operation
  - Not very accurate

Table 10.2 Recommended Number of Observation Cycles

Cycle time (min)	Recommended number of cycles
0.10	200
0.25	100
0.50	60
0.75	40
1.00	30
2.00	20
2.00–5.00	15
5.00–10.00	10
10.00–20.00	8
20.00–40.00	5
40.00–above	3

Source: Information taken from the Time Study Manual of the Erie Works of the General Electric Company, developed under the guidance of Albert E. Shaw, manager of wage administration.

Number of cycles

- Statistical techniques are more accurate
- Need previous estimates, predictions, or a few sample calculations
  - Pilot study: record a sample of cycles (< 30) before starting the actual experiment
  - Then estimate the number of cycles n based on the element times of that sample
  - If there are multiple elements, take the highest n
  - Confidence level (CL): typically 90, 95, or 99%
- Number of cycles (sample size) is typically less than 30, so use the t-statistic confidence interval:
  - Where
    - $\bar{X}$ is the sample mean
    - t is the t-score
    - S is the standard deviation of the sample
    - n is the desired number of cycles

$$\bar{X} \pm t\frac{S}{\sqrt{n}}$$

Margin of error

Number of cycles

- t can be found when you know the significance α and degrees of freedom $(n-1)$

Degrees of freedom = current sample size - 1

	Probability P												
n	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1	0.05	0.02	0.01	0.001
1	0.158	0.325	0.510	0.727	1.000	1.376	1.963	3.078	6.314	12.706	31.821	63.657	636.619
2	0.142	0.289	0.445	0.617	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	31.598
3	0.137	0.277	0.424	0.584	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	12.941
4	0.134	0.271	0.414	0.569	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	8.610
5	0.132	0.267	0.408	0.559	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	6.859
6	0.131	0.265	0.404	0.553	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.959
7	0.130	0.263	0.402	0.549	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	5.405
8	0.130	0.262	0.399	0.546	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	5.041
9	0.129	0.261	0.398	0.543	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.781
10	0.129	0.260	0.397	0.542	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.587
11	0.129	0.260	0.396	0.540	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.437
12	0.128	0.259	0.395	0.539	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	4.318
13	0.128	0.259	0.394	0.538	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	4.221
14	0.128	0.258	0.393	0.537	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	4.140
15	0.128	0.258	0.393	0.536	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	4.073
16	0.128	0.258	0.392	0.535	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	4.015
17	0.128	0.257	0.392	0.534	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.965
18	0.127	0.257	0.392	0.534	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.922
19	0.127	0.257	0.391	0.533	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.883
20	0.127	0.257	0.391	0.533	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.850
21	0.127	0.257	0.391	0.532	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.819
22	0.127	0.256	0.390	0.532	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.792
23	0.127	0.256	0.390	0.532	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.767
24	0.127	0.256	0.390	0.531	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.745
25	0.127	0.256	0.390	0.531	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.725
26	0.127	0.256	0.390	0.531	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.707
27	0.127	0.256	0.389	0.531	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.690
28	0.127	0.256	0.389	0.530	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.674
29	0.127	0.256	0.389	0.530	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.659

Two-sided alpha (given)

t values (that you find)

Appendix A3.3

Number of cycles

- The margin of error can be rewritten as a function of $\bar{X}$ using k, the *acceptable fraction* of $\bar{X}$

$$\bar{X} \pm t\frac{S}{\sqrt{n}}$$

$$k\bar{X} \pm t\frac{S}{\sqrt{n}}$$

- The acceptable number of cycles n is then (always round up): $n = (\frac{tS}{k\bar{X}})^2$

- **<u>Example 2</u>**: Assume you had 25 observations of a work element, and the mean of those readings is 0.3 minutes and the standard deviation is 0.09 minutes. If the acceptable fraction for the error term is 5%, and the desired alpha is 0.05, calculate the targeted number of observations using the statistical technique

Example 3

- What is the recommended number of cycles based on this pilot study?

n	Element 1 time (min)
1	0.481
2	0.465
3	0.519
4	0.537
5	0.596
6	0.525

- n = 6
- $\bar{X} = 0.52 \text{ min}$
- S = 0.046 min
- K = 0.05
- α = 0.05 = 1 - 0.95
- Degrees of freedom = 5
- t = 2.571
- $n = (\frac{tS}{k\bar{X}})^2$= 21 observations

Given CL = 0.95; k = 0.05

Example 4 - What if there is more than 1 element?

- What is the recommended number of cycles based on this pilot study?

n	Element 1 time (min)	Element 2 time (min)
1	0.481	0.622
2	0.465	0.610
3	0.519	0.610
4	0.537	0.598
5	0.596	0.640
6	0.525	0.611

Given CL = 0.95; k = 0.05

- n = 6
- $\bar{X} = 0.674$ min
- S = 0.046 min
- K = 0.05
- α = 0.05 = 1 - 0.95
- Degrees of freedom = 5
- t = 2.571
- $n = (\frac{tS}{k\bar{X}})^2$= 2 observations
- Always take the largest n = 21

Time study procedure

1. Identify breakpoints for your elements
2. Determine number of cycles needed
3. Record the element times
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5. Calculate the elemental times
6. Performance rate the operator
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Record element times

- Set up your time study form
- Record the times in person or using a video for all the cycles you need
  - If this is a pilot study, use the preliminary data to establish n
- Two methods can be used to record element times:
  - Snapback technique: Watch is reset back to zero after each recording
  - Continuous timing

	Element 1	Element 2	Element 3
Cycle 1			
Cycle 2			
Cycle 3			
Cycle 4			
Cycle 5			
Cycle 6			

- Only normal elements go in the table

Indicate the unit of time

Time study procedure

1. Identify breakpoints for your elements
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Dealing with abnormal elements

- Indicate abnormal elements that occur with a circled label (A, B, etc.), place it where it occurred, and explain in the footnotes
- Case 1: If it occurred in between two normal elements, record in the footnotes for completeness and ignore
- Case 2: If it occurred as part of a normal element
  - Case 2a: If the abnormal element is long enough, record its time separately below the table and later subtract it from the element time
  - Case 2b: If it is not, just record it as part of the normal element time and circle that element time. Then, we determine if the time is an outlier

	Element 1	Element 2	Element 3
Cycle 1	A		
Cycle 2			
Cycle 3		B	
Cycle 4			

A: operator talking to another worker
B: operator answering phone call

Example 5

- Is 6 an outlier?

Element
3
3
2
4
5
6 C
4
4
4

Q1 = 3
Q3 = 4
(inclusive)

- IQR = 1
- 1.5*IQR = 1.5
- 6>Q3+1.5IQR= 5.5

So 6 is an outlier; remove it from calculations

Element
3
3
2
4
5
- C
4
4
4

C: operator dropped the hammer

Time study procedure

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Calculate elemental times

- Find the mean elemental times for each element
- Find the standard deviation for each elemental time

	Element 1	Element 2	Element 3
Cycle 1			
Cycle 2			
Cycle 3			
Cycle 4			
Cycle 5			
Cycle 6			
Average	$\overline{ET_1}$	$\overline{ET_2}$	$\overline{ET_3}$
Standard dev	σ_1	σ_2	σ_3

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